Reg. No. | | | | | | | | | | |

[16CSS01604]

SIXTH SEMESTER B.TECH UNIVERSITY EXAMINATION, JUNE - 2019

PYTHON PROGRAMMING

Time: 2 Hours **Maximum Marks:40**

Instructions:

1. Answer any **FIVE** full Questions selecting at least two full questions from each part.
2. Missing data may be suitably assumed.

PART-A

1. a. Discuss the usage of the following with respect to the print() function with an example. **04**

i. sep argument **ii.** end argument

b. Write a Python program to find the best of two test average marks out of three test's marks accepted from the user. **04**

2. a. With a suitable example, explain variable number of arguments in Python. **04**

b. Write a user defined function named "Compute" that calculate the Remainder and Quotient on division of two numbers accepted from the user. Print the remainder and quotient separately on the console. **04**

3. a. Design an interactive guessing game, which will ask the user to guess a number between 1 and 99. **03**

b. What is the output of the following program? Draw a stack diagram that shows the state of the program when it prints the result. **05**

```
def recurse(n, s):
    if n == 0:
        print(s)
    else:
        recurse(n-1, n+s)
recurse(3, 0)
```

i. What would happen if you called this function like this: recurse(-1, 0)?

ii. Write a docstring that explains everything someone would need to know in order to use this function (and nothing else).

4. a. Write a Python program to get a string made of the first 2 and the last 2 characters from a given a string. If the string length is less than 2, return empty string. **04**

b Using string method, write an expression that produces: **04**

i. The number of 'a's in graduate.

ii. The index of first occurrence of 'a' in graduate.

iii. A copy of 'graduate' capitalized.

iv. Copy of "Graduate" with leading white spaces removed.

Also explain the method used in each operation.

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PART-B

5. a. Consider the list li=[10,5,6,19,26,32,50]. Give the output of the following commands. **04**

   i. li[0:-1] ii. li[:] iii. li[::-1] iv. li[-1:-4]

   b. Compare list and tuple with suitable examples. **04**

6. a. Write a Python program to check whether a given key exists in a dictionary or not. **04**

   b. Describe the process of storing and accessing data using dictionaries. **04**

7. a. Write a Python program to remove newline characters from a file. **04**

   b. Describe the different directory management functions used in Python. **04**

8. a. Write a function called sed that takes as arguments a pattern string, a replacement string and two filenames. It should read the first file and write the contents into the second file (creating it if necessary). If the pattern string appears anywhere in the file, it should be replaced with the replacement string. **05**

   If an error occurs while opening, reading or closing files, the program should catch the exception, print an error message and exit.

   b. With a suitable programming example, explain how exceptions are handled in Python. **03**